

MO I RANA, Norway

WMO: 011510

Lat: 66.364N

Lon: 14.301E

Elev: 230

StdP: 14.57

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -9.2	(c) -3.7	(d) -13.7	(e) 2.6	(f) -7.6	(g) -9.2	(h) 3.4	(i) -2.4	(j) 26.8	(k) 29.3	(l) 22.8	(m) 28.5	(n) 2.5	(o) 330	(p) 0.676

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.7	78.1	61.8	73.4	59.6	69.6	57.7	63.4	75.0	60.8	70.9	58.7	67.5	7.5	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
58.6	74.2	66.4	56.3	68.1	63.7	54.5	63.8	61.8	28.8	75.5	27.0	70.8	25.6	67.6	72.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.0	16.6	14.5	-13.3	83.6	4.5	5.6	-16.6	87.6	-19.2	90.9	-21.7	94.1	-24.9	98.2
			-14.3	65.9	4.5	3.8	-17.5	68.7	-20.1	70.9	-22.6	73.0	-25.8	75.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	38.0	20.7	22.0	26.1	34.5	44.2	51.1	58.2	55.5	49.1	38.1	29.3	25.8
	DBStd	15.20	12.08	11.70	8.80	6.01	6.82	5.72	6.72	5.03	5.76	6.50	8.61	9.93
	HDD50	4997	909	785	741	465	211	52	2	8	84	370	620	750
	HDD65	9887	1374	1205	1206	915	644	419	233	297	476	833	1070	1215
	CDD50	611	0	0	0	0	32	84	256	179	58	2	0	0
	CDD65	27	0	0	0	0	0	1	23	3	0	0	0	0
	CDH74	343	0	0	0	0	14	10	282	37	0	0	0	0
	CDH80	65	0	0	0	0	1	0	63	1	0	0	0	0
Wind	WSAvg	5.7	6.1	5.9	6.1	5.5	6.5	6.3	5.3	5.1	5.1	5.3	5.2	5.5
Precipitation	PrecAvg	57.5	6.1	5.7	5.7	3.4	2.7	3.0	3.4	3.6	5.7	6.4	5.2	6.7
	PrecMax	75.7	15.0	11.9	11.3	8.5	4.5	6.1	8.0	6.9	17.5	14.5	10.3	15.4
	PrecMin	39.3	0.2	0.4	1.4	0.7	1.1	0.5	1.4	0.9	1.3	1.4	0.4	0.9
	PrecStd	8.9	3.9	3.1	3.0	2.0	0.9	1.7	1.5	1.7	3.5	3.0	3.2	4.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.7	42.2	44.2	59.5	74.5	76.9	86.1	78.4	69.2	57.5	48.5	43.4
		MCWB	41.4	39.8	38.5	46.8	56.0	60.1	66.6	61.4	58.2	50.0	44.2	40.4
	2%	DB	41.0	40.2	41.3	50.6	68.8	71.1	80.7	74.0	64.6	53.7	45.9	41.1
		MCWB	38.4	37.3	37.5	41.0	53.7	57.1	63.1	60.5	54.8	47.0	42.7	36.7
	5%	DB	38.5	37.9	39.4	46.3	63.5	67.3	76.5	70.1	61.6	49.9	43.2	39.0
		MCWB	35.7	35.0	35.8	39.4	51.4	54.8	62.0	59.1	53.6	45.6	39.7	35.3
	10%	DB	35.6	35.7	37.2	43.5	56.8	63.5	71.8	66.4	58.6	46.8	40.3	36.8
		MCWB	33.1	33.1	33.8	37.7	47.2	52.9	59.3	57.1	52.1	43.7	37.4	33.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.4	40.6	40.6	46.8	57.1	61.1	69.1	64.2	59.3	50.9	45.6	41.8
		MCDB	43.3	41.8	42.3	57.7	73.0	72.2	81.7	73.2	65.7	56.6	47.2	42.6
	2%	WB	38.8	37.9	38.5	42.9	54.3	58.0	65.4	61.3	56.5	48.0	43.0	37.6
		MCDB	40.0	39.5	40.8	47.2	67.4	69.0	78.5	71.6	62.5	51.9	45.2	39.6
	5%	WB	35.7	35.4	36.2	40.4	51.4	55.8	62.5	59.4	54.7	46.0	40.1	35.6
		MCDB	37.9	37.2	38.5	44.6	63.3	65.2	74.0	68.4	60.1	48.8	42.8	37.9
	10%	WB	33.2	33.4	34.2	38.5	48.1	53.5	60.2	57.7	52.8	44.2	37.3	34.2
		MCDB	34.7	35.2	36.4	42.5	55.3	61.6	70.0	65.1	57.7	46.5	40.0	36.4
Mean Daily Temperature Range	5% DB	MDBR	11.3	11.9	14.2	14.1	15.4	14.7	15.7	15.6	14.1	9.9	8.4	10.4
		MCDBR	9.9	9.9	12.3	19.8	26.9	23.3	25.7	23.8	18.7	12.6	9.8	10.8
		MCWBR	8.6	8.4	9.3	12.6	14.0	12.0	11.4	12.1	11.0	8.8	7.8	9.0
	5% WB	MCDBR	10.8	9.6	10.7	15.7	25.1	20.5	22.5	21.6	16.6	10.2	9.1	9.7
		MCWBR	9.9	8.7	8.6	10.7	13.3	10.9	10.6	11.5	10.4	7.9	7.8	8.2
Clear-Sky Solar Irradiance	taub	0.203	0.270	0.307	0.335	0.342	0.342	0.353	0.339	0.316	0.285	N/A	0.209	
	taud	1.920	2.103	2.134	2.169	2.263	2.381	2.424	2.465	2.506	2.284	N/A	1.933	
	Ebn at Noon	64	180	233	259	271	275	267	260	240	178	N/A	5	
	Edh at Noon	5	17	28	35	36	32	30	26	20	14	N/A	1	
All-Sky Solar Radiation	RadAvg	21	141	552	1134	1521	1568	1424	1076	571	205	44	3	
	RadStd	2	27	79	97	92	153	167	115	68	35	4	0	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (25 neighbors)	+0.67	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page